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Heine-Borel without sequential compactness

(Exercise 3.3.10 in Abbott)

Definition 1: A set is *compact* if every open cover of it has a finite subcover. Note that this is different from how Abbott defines compactness in terms of convergent subsequences.

We make this definition to avoid clunky repetition of the phrase “a set with the property that every open cover has a finite subcover.”

Let $[a, b]$ be any closed interval in $\mathbb{R}$. Let $\{O_\lambda : \lambda \in \Lambda\}$ be an open cover for $[a, b]$. Define S to be the set of all $x \in [a, b]$ such that $[a, x]$ has a finite subcover from $\{O_\lambda : \lambda \in \Lambda\}$.

Lemma 1: $\sup S$ exists and satisfies $a \leq \sup S \leq b$.

Proof: Since $\{O_\lambda : \lambda \in \Lambda\}$ covers $[a, b]$ and $a \in [a, b]$, there must be at least one $O_a \in \{O_\lambda\}$ that covers a . Then the singleton set $\{O_a\}$ constitutes a finite subcover of $[a, a]$. So $a \in S$.

By the definition of S , it consists of points $x \in [a, b]$ with a certain property, therefore it is a subset of $[a, b]$. So b is an upper bound for S .

Since $S \subseteq \mathbb{R}$ is nonempty and bounded, it has a least upper bound.

Since $a \in S$ and the supremum is an upper bound, $\sup S \geq a$.

Since every $x \in S$ satisfies $x \leq b$, by an argument from one or two problem sets ago, $\sup S \leq b$. Combining the two inequalities gives the desired result. ■

Lemma 2: If $x \in S$ and $a \leq y \leq x$ then $y \in S$.

Proof: Since $x \in S$, there's a finite subcover of $[a, x]$ from $\{O_\lambda : \lambda \in \Lambda\}$. This finite subcover also covers $[a, y]$ since $[a, y] \subseteq [a, x]$. So $y \in S$. ■

Lemma 3: $\sup S = b$.

Proof: Let $s = \sup S$.

By the lemma above, $s \in [a, b]$, so there must be at least one $O_s \in \{O_\lambda : \lambda \in \Lambda\}$ that covers it. Since O_s is open in $\mathbb{R}$, there's some ball $B_\varepsilon(s)$ centered at s with radius ε contained entirely inside O_s .

We may take $\varepsilon < b - s$ in the rest of the argument.

Suppose for the sake of contradiction that $s < b$. Then let

$$t = s + \frac{\varepsilon}{2}$$

so t satisfies

$$s < t < b \text{ and } t \in B_\varepsilon(s).$$

Then $t \in O_s$ as well, since the whole ball is contained in O_s .

By the definition of the supremum, there's some $y \in S$ with $y > s - \frac{\varepsilon}{2}$. So there's a finite subcover Φ of $[a, y]$.

By adjoining O_s to Φ we have a finite subcover of $[a, t]$, so $t \in S$ but $t > s$, a contradiction of s being an upper bound.

So $s \leq b$ but we have shown that $s < b$ is impossible. So $s = b$. ■

Lemma 4: $[a, b]$ has a finite subcover.

Proof: There must be at least one $O_b \in \{O_\lambda : \lambda \in \Lambda\}$ that covers b . Since O_b is open, there's some ball $B_\varepsilon(b)$ centered at b with radius ε contained entirely inside O_b . We may take $\varepsilon < b - a$, for, if $b = a$ then $[a, a]$ is trivially finitely subcoverable and there is nothing more to prove. Since $b = \sup S$, by the definition of the supremum and of the set S , there's a finite subcover Φ of $[a, b - \frac{\varepsilon}{2}]$. By adjoining O_b to Φ , we have a finite subcover of $[a, b]$. ■

Several problem sets ago we proved the following statement for sequential compactness. Here's the proof for the definition of compactness based on open covers:

Lemma 5: Any closed subset of a compact set is also compact.

Proof: Let $K \subseteq \mathbb{R}$ be compact and $L \subseteq K$ a closed subset. Then L^C is open in $\mathbb{R}$. Let C be an open cover of L . Then $C \cup \{L^C\}$ is an open cover of all of $\mathbb{R}$, and in particular of K . By compactness of K , there's a finite subcover Φ of K from $C \cup \{L^C\}$.

Since $L \subseteq K$, Φ covers L as well. If $L^C \notin \Phi$ then $\Phi \subseteq C$ and is the desired finite subcover of L from C . Otherwise, if $L^C \in \Phi$, then $\Phi \setminus \{L^C\}$ still covers L and is still finite, since L^C could not have been used to cover any of L . Either way we have found a finite subcover of L . ■

Theorem 1: Let K be any closed and bounded subset of $\mathbb{R}$. Then K is compact.

Proof: Since K is bounded it is contained in some interval $[-M, M]$. The interval is compact, and K is a closed subset of it, so K is compact. ■

Composition of continuous functions

Theorem 2: Let $f : R^m \rightarrow R^n$ and $g : R^l \rightarrow R^m$ be continuous. Then $f \circ g : R^l \rightarrow R^n$ is continuous.

Proof: Let $N \subseteq R^n$ be closed. Then its preimage M under f is a closed subset of R^m . And the preimage of M under g (call it L) is a closed subset of R^l .

That is, the preimage under $f \circ g$ of any closed $N \subseteq R^n$ is closed. So $f \circ g$ is continuous. ■